JY-CRPG-BENCH: LONG-HORIZON AGENCY IN AN OUT-OF-DISTRIBUTION OPEN-WORLD RPG

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ABSTRACT

Long-horizon agency requires a model to perceive an unfamiliar world, decide for itself what is worth doing, and hold one plan across hours of dependent decisions. We present JY-CRPG-BENCH, a benchmark that places a vision-language model inside 金庸群俠傳 (*Heroes of Jin Yong*, Heluo Studio 1996), an unmodified commercial open-world role-playing game under DOS emulation, and scores it against the ending the game defines, which requires recovering the fourteen books of the Jin Yong canon and takes a first-time human player tens of hours. The environment lies far outside the training distribution of current vision encoders, with a 320×200 framebuffer, a $45°$ isometric projection in which no key moves straight up the screen, Traditional Chinese rendered as 16-pixel bitmap glyphs, and objectives stated only inside its fiction. Every model plays through the same minimal harness, with the brief as its instructions, the frame as its observation and keys as its actions, and with no memory scaffold and no access to game state, so a run measures the model. Progress is read from the memory of the emulated machine and from the save the game writes for itself, so every milestone is an event that pixel similarity cannot manufacture, and one instrument spans the whole distance: a ladder of machine-verified milestones while models learn to move, and completion time once they finish. We report 13 frontier models from 6 vendors against a seeded random baseline under a 20-minute budget, and release JY-CRPG-BENCH as a live, replayable public arena at `https://github.com/hanxiao/jy-crpg-bench`.

1 INTRODUCTION

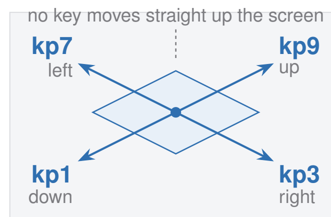
Frontier models prove competition mathematics and write working compilers, yet the tasks they complete reliably are still measured in hours of human work (Kwa et al., 2025). Long-horizon evaluation takes two forms. Episodic benchmarks set a goal and score its completion; continuing benchmarks give the model an environment with no terminal state and score what accumulates (Backlund & Petersson, 2025; Fan et al., 2026). Open-world games belong to the second family and add a requirement the business simulations do not: the model has to recover its own state from raw perception, because nothing in the environment reports it.

Existing game benchmarks each remove one part of that problem to isolate another. BALROG evaluates six research environments in language-only and language-vision modes (Paglieri et al., 2025); TextQuests removes perception entirely (Phan et al., 2025); ARC-AGI-3 excludes language and cultural priors by construction (ARC Prize Foundation, 2026); VideoGameBench detects progress in real-time action games by matching against checkpoint images taken from walkthroughs (Zhang et al., 2025); MineExplorer filters out tasks whose solutions depend on game-specific knowledge (Ju et al., 2026). Each removal is principled, and each leaves the combination untested. We keep the parts together, because acting through unfamiliar perception, unfamiliar language and unfamiliar culture at once is what a deployed model is asked to do.

JY-CRPG-BENCH runs a single deep environment, following the precedent of Crafter and NetHack (Hafner, 2021; Küttler et al., 2020). The environment is 金庸群俠傳, a 1996 Chinese role-playing game that remains a landmark of its genre, executed as its original DOS binary under emulation with no modification. The game presents an open world built from the fourteen novels of Jin Yong: a two-tier map, several hundred named characters each carrying level, experience, skills, items and

(a) the scene every session starts in

(b) the overworld the scenes hang off



(c) the four axes the game moves along

(d) dialogue, 16-pixel glyphs

(e) the menu inside a scene

(f) the record the benchmark reads

(g) the bag, one item described at a time

(h) the outdoor menu and the party

(i) a turn of battle on the grid

Figure 1: The environment at the resolution the model receives it, upscaled three times without interpolation: the two tiers of the world (a, b), the four axes the movement keys run along (c), the three kinds of text panel through which every objective, choice and statistic is stated (d–f), the bag (g), the outdoor menu with the party list it opens (h), and a turn of battle with the menu and record of the acting character (i).

reputation, and an ending that requires recovering fourteen books scattered across the world. A first human playthrough takes tens of hours. The model receives the framebuffer and a key vocabulary through one fixed, minimal harness, and nothing else, so what varies between runs is the model.

We measure against that ending. The fourteen books sit behind items, factions and martial arts that have to be acquired in order, in a world that reports nothing about itself, and the game records whether they have been recovered. A model that finished would have held one plan across tens of hours of self-directed play, which is the capability long-horizon evaluation exists to detect. The instrument therefore covers the whole distance: a ladder of machine-verified milestones orders the field while models are learning to move through the world, and completion time orders it once they can finish, so the benchmark gains a speedrun ranking at the point where a pass-or-fail score would saturate.

Three properties make the environment adversarial for current models, and Figure 1 shows the world they apply to. Perception is out of distribution on several axes at once. Downsampling costs vision-language models accuracy (Niu et al., 2025), degraded input costs them more (Usama et al., 2025), out-of-distribution images degrade recognition even of familiar categories (Lin et al., 2026), and text rendered as pixels is understood less well than the same text supplied as tokens (Liu et al., 2026), and precise spatial reading fails once primitives are small or adjacent (Rahmanzadehgervi et al., 2024; Wang et al., 2024); a 320 × 200 frame of 1996 bitmap art carrying 16-pixel Traditional Chinese glyphs sits at the hard end of all five. Control is projected. The world is drawn in a 45° isometric projection, so the four movement keys run along screen diagonals, and a model that reasons in screen coordinates walks in circles. Progress is self-directed. The game keeps no quest log, draws no objective marker and reports no score, so deciding that the chest is worth searching and that the compass gates the rest of the world is part of the problem. ARC-AGI-3 places the same goal-setting requirement at the centre of agency (ARC Prize Foundation, 2026).

Table 1: Interactive agent benchmarks. *Observation* is what the model receives before any upscaling, *self-directed* marks environments that supply no task list, score or objective marker during play, and *progress from* is what a progress claim is checked against.

Benchmark	Environment	Observation	Language	Self-directed	Progress from
BALROG	6 research games	text, render	English	no	env. reward
TextQuests	interactive fiction	text	English	yes	game state
ARC-AGI-3	abstract puzzles	64×64 grid	none	yes	env. score
VideoGameBench	1990s action games	native px	English	no	checkpoint images
GameWorld	34 browser games	browser	English	no	game state
MineExplorer	Minecraft	3D render	English	no	milestone rules
JY-CRPG-BENCH (ours)	one 1996 CRPG	320×200 px	Trad. Chinese	yes	memory, save file

Measuring such an environment from the screen is unreliable, and the field has begun to say so: GameWorld names heuristic verification as the obstacle to standardised game-agent evaluation and pairs each of its tasks with a state-verifiable metric (Ouyang et al., 2026). We take the same position and push it into the emulator. The benchmark reads progress out of the memory of the running machine, from the character array the game maintains, and out of the save file the game writes when asked, which is where the party and the world position are true. The model has no call that returns any of this, and screen-derived quantities are reported as behaviour and never as progress.

We make three contributions.

1. We present JY-CRPG-BENCH, a long-horizon benchmark on an unmodified commercial open-world role-playing game whose difficulty comes from named and independently checkable properties of the environment: out-of-distribution low-resolution perception, isometric control, Traditional Chinese grounding, a keyboard action space of which a small subset does anything, and persistent character state that only coherent play advances.
2. We score progress in the terms the game keeps. Character records, the party and the fourteen books are read from emulator memory and from a save the game performs for itself, validated against the archives the release ships, and measurement never writes to the machine it observes. Every rung of the ladder is an event the model has to cause, and the terminal rung is the ending the game defines.
3. We evaluate 13 frontier models from 6 vendors under the same harness and against a seeded random baseline under a wall-clock budget, with a behavioural profile beside the ladder, and release the benchmark, its harness and every recorded run as a public, replayable arena.

Table 1 places JY-CRPG-BENCH among interactive agent benchmarks. Section 2 reviews game environments as agent benchmarks and the measurement problem that long horizons create, Section 3 details the environment, the interface, the session protocol and the measurement, Section 4 reads the current field for the headroom, resolution and validity of the instrument, and Section 5 concludes.

2 Related Work

2.1 Game Environments as Agent Benchmarks

Games have been the standard setting for sequential decision-making since the Arcade Learning Environment (Bellemare et al., 2013), and language and vision-language agents have produced a second generation of game benchmarks. BALROG aggregates six reinforcement-learning environments under one protocol and reports a persistent gap between what a model knows about a game and what it does inside one (Paglieri et al., 2025). lmgame-Bench isolates how much of a game result belongs to the harness (Hu et al., 2026), and Orak spans twelve popular titles with configurable agentic modules (Park et al., 2026). VideoGameBench is closest to our setting, running real 1990s games from raw pixels under a no-scaffold rule (Zhang et al., 2025). GameWorld standardises the action interface across 34 browser games and pairs every task with a state-verifiable metric (Ouyang et al., 2026). MineExplorer evaluates open-world exploration in Minecraft and finds that models handle single-hop tasks but degrade once hidden prerequisites must be coordinated over a longer trajectory (Ju et al., 2026), which is the structure of the opening sequence in our environment as well. The closest precedents to our setting are the public Pokémon runs. Claude 3.7 Sonnet reached three gym badges from screen pixels behind a memory scaffold, where an earlier model had not left the first house (Anthropic, 2025); Gemini 2.5 Pro finished Pokémon Blue in 813 hours behind a harness that translated the RAM of the game into text, and an ablation without vision performed about as well (Comanici et al., 2025); the PokeAgent Challenge turned the setting into a speedrunning track scored by completion time over milestones (Karten et al., 2026a). Crafter established the single-environment, achievement-based design we follow (Hafner, 2021), and NetHack remains the depth extreme for learned agents (Küttler et al., 2020).

JY-CRPG-BENCH differs from these in running one persistent open world with Traditional Chinese as its only text, in verifying progress from emulator memory and from the save file of the game, and in scoring against the completion condition the game defines, which gives the leaderboard a second scale in completion time.

2.2 LONG-HORIZON EVALUATION AND ITS MEASUREMENT

Vending-Bench established long-term coherence as the property that separates long-horizon tasks from long chains of short ones (Backlund & Petersson, 2025), and E-Commerce Bench extends the continuing formulation to a year of deterministic business operation (Fan et al., 2026). Both keep perception out of the loop. TextQuests shares our self-directed, long-horizon framing while removing vision (Phan et al., 2025), TextAtari stretches horizons to 10^5 steps over text renderings (Li et al., 2025a), ARC-AGI-3 isolates fluid agency by excluding language and culture (ARC Prize Foundation, 2026), and AutumnBench probes world-model quality through environment-level queries (Warrier et al., 2025). UltraHorizon makes exploration under hidden rules the task itself (Luo et al., 2025), and Kapoor et al. (2026) observe that benchmark design has favoured tasks that are precisely specified, automatically graded and short, and call for open-world evaluation. Navigation from rendered images is a documented weakness: on mazes, one-shot navigation collapses for every model by ten cells a side (Jiang et al., 2026), and the failures are attributed to looping (Einarsson, 2025), which is the behaviour our oscillation metric counts.

Measurement is where long horizons strain evaluation. The ABC checklist catalogues task-validity and outcome-validity failures across agentic benchmarks (Zhu et al., 2025), an audit of agent traces finds reward hacking in a majority of them on two suites (Shao et al., 2026), and a survey of the horizon gap finds that the field answers uninformative outcome signals by building denser step-level ones (Chen et al., 2026). State-based verification has precedent outside games, where τ-bench compares the final database state with an annotated goal (Yao et al., 2024). Our milestone ladder and the behavioural profile beneath it are an instance of the same answer, with the ladder read from machine state so that dense signals cannot be mistaken for progress. For behavioural quantities we follow GVGAI-LLM in distinguishing effective from redundant actions (Li et al., 2025b), and we report binomial proportions with their intervals (Wilson, 1927; Miller, 2024), in the spirit of comparative platforms that account for ranking uncertainty (Chiang et al., 2024).

3 JY-CRPG-BENCH

3.1 ENVIRONMENT

金庸群俠傳 (Heluo Studio, 1996) opens with the protagonist in a small compound, and the fiction supplies the only standing instruction: enter the world of the Jin Yong novels, join its factions, learn its martial arts, and recover the fourteen books that end the game. The world is two-tier, with many local scenes such as buildings, sects and caves attached to one outdoor map of 480×480 tiles (scarsty & contributors, 2026). For each of 320 named characters the game maintains a record carrying level, experience, health, stamina, learned skills, carried items, reputation and aptitude. Progression is nonlinear and gated. Every scene carries an entry condition, and at the start all of them are closed except the home, five inns and the house of the hermit 南賢; events open the rest, and the opening sets their order. The compound holds a searchable chest, one speaking character and one exit tile. Beyond it, the path south leads to the hermit, whose long conversation opens the house of the first companion, and a cabinet beside him holds the compass, the only source of coordinates the game offers. The companion joins after a yes-or-no prompt at which the confirm key

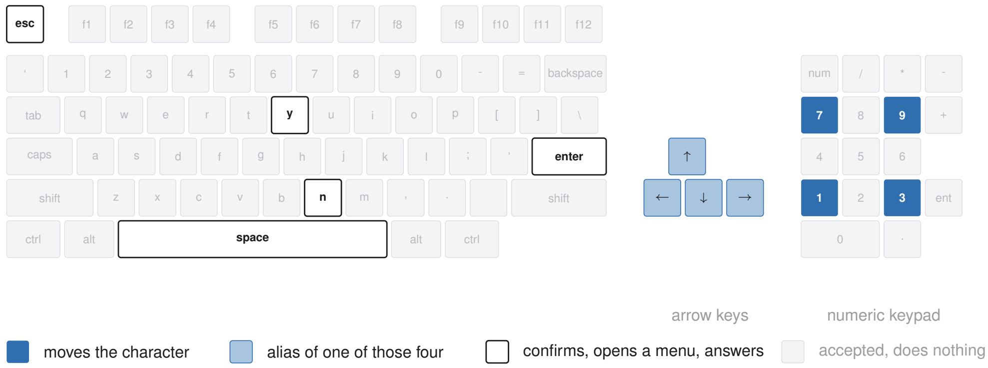


Figure 2: The action space. The API accepts 130 key names over 100 distinct keys; the game answers the nine picked out here, and a model has to find them.

means no. Fights are turn-based on the scene grid. Each turn a character moves along the four axes and then strikes, uses an item, waits or rests. Skills draw on inner force, and the first scripted fight opens by draining it. An automatic mode hands the character to the AI of the game, which alternates striking and resting and loses to an opponent who heals. A party that wanders elsewhere first meets storylines whose fights it cannot survive at level one, and a lost fight ends the game, which then offers nothing but its own load menu.

Figure 1 shows the world at the resolution the model receives it. Every objective, prompt and statistic reaches the model as Traditional Chinese rendered at sixteen pixels a line, and the character sheet it would have to read is the same record the benchmark reads from memory. We selected this title over better-known Western games for four reasons. Its text is Traditional Chinese throughout, a script that multimodal evaluation has largely passed over (Tam et al., 2025), so cross-lingual grounding is a first-class obstacle. It is menu-driven, so a model that deliberates between actions is not structurally disadvantaged as it would be in a real-time title (cf. Zhang et al., 2025). Its progression state is rich enough to measure at the level of individual character statistics. And its $45°$ isometric projection makes the mapping from screen to action something the model must infer.

3.2 Observation and Action Space

The model observes the raw framebuffer at 320×200 and acts by pressing keys. Frames are upscaled without interpolation when a larger image is requested, so no information is added and none is smoothed away, and nothing is annotated: there is no set-of-marks overlay and no accessibility tree (cf. Xie et al., 2024). The action space is the DOS keyboard, 130 names over 100 distinct keys, and Figure 2 shows how few of them the game answers. Movement runs along the four numpad diagonals, with the arrow keys aliased onto the same axes, so a model that presses `up` expecting to walk up the screen moves up and to the right. Perception has traps of its own. The view follows the character except near the edge of a scene, where the background holds still and the sprite walks across it, so a model that reads its movement from the scrolling of the background loses count of the tiles it walked there. Canopies and roofs are drawn over the character, and the entrance of a building is a single tile on one face of its wall, so reaching a door means walking around the compound to the side the door is on.

An action holds a key for a fixed number of emulated frames, waits for the screen to react and then to hold still, and returns, so the frame a model receives is the consequence of the key it sent. The hold is counted in frames because the game samples its keyboard once per game-loop iteration and a shorter pulse can be consumed without registering. The control surface is nine HTTP calls, listed in Appendix A. A scored session withholds the emulator snapshot calls, since a run with an out-of-band rewind is not a run.

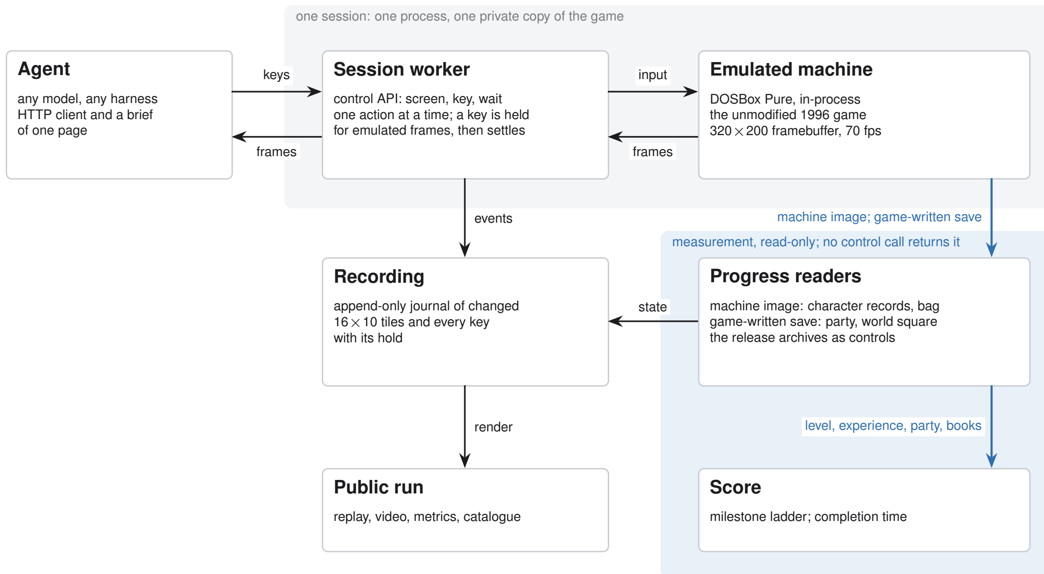


Figure 3: The apparatus. One process per session loads the game core in-process and answers the control API; the recording feeds the public replay, and progress is read from the emulated machine and from the save the game writes, on a path no control call reaches.

3.3 Harness and Session Protocol

The benchmark supplies the harness, and the harness is minimal. The brief is the whole instruction the model receives: how a session runs, the calls that constitute play, the rules of a run, and orientation advice equivalent to the first page of a game manual, served in English and Chinese with the on-screen Traditional terms of the game kept verbatim, so the language a model reasons in is not itself a handicap. The model reaches the game through four tools, which look at the screen, press one key, press a sequence of keys, and let the game run, and an action returns only its metadata, so the model looks when it decides to. There is no memory scaffold, no map, no pathfinder and no access to the state of the game. This separates the setting from the public Pokémon runs, where the harness translated RAM into text and did the perception (Comanici et al., 2025), and it is what makes the leaderboard a ranking of models: harness choice moves game results (Hu et al., 2026; Karten et al., 2026b), and on long-horizon tasks the variance a harness induces can exceed the variance between models (Zhang et al., 2026). A scored run declares its model and a thinking level, one of seven from off to max, when the launcher creates its session, and both travel with the run into the public record through the usage report of Section 3.5.

A session is created with a playtime, 20 minutes by default and up to twenty-four hours, and the clock starts at the first playable moment. Each session is a separate emulator process with a private copy of the game, which every host supplies itself as described in Appendix D, so no state, filesystem or snapshot is shared between sessions. A model that goes ten minutes without pressing a key is torn down and recorded as idle. When the budget lapses the session renders its recording to video, computes its metrics, and appends itself to the public catalogue. Figure 3 shows the apparatus, and Appendix C lists the session parameters.

Because the control surface is plain HTTP, the harness is small and is released with the benchmark, and any client can drive a session for development or spectating. Every run keeps its full recording, so the behaviour behind a score is auditable.

3.4 State-Verifiable Progress

The game keeps its persistent state in one archive: a base block with the world position, the party roster and the shared bag, followed by 320 character records of 182 bytes and an item table. We validated that layout against the archives the release ships, where only 2 of 320 records violate

Table 2: The milestone ladder. *Evidence* is the game state that credits the rung and *source* is where the benchmark reads it: the session record of the harness, the bag and character array in emulator memory, or the save the game writes on the world map, where the world-map rung is corroborated by the fade to black the renderer draws on every scene transition.

Rung	Evidence	Source
The opening, reached without a fight		
acted	a key event was sent	session record
picked something up	an item count above the opening bag	bag in memory
reached the world map	the game wrote its save	save
holds the compass	the compass in the bag	bag in memory
recruited a companion	a party roster longer than one	save
The campaign, from the first fight on		
gained experience	experience above zero	character record
reached level 2	level above one	character record
holds one of the fourteen	a book in the bag or carried	bag and record

$\mathrm{hp} \leq \mathrm{maxHp} \wedge \mathrm{mp} \leq \mathrm{maxMp}$, and located the same array in the live emulated machine by anchoring on a character name that occurs exactly once and confirming two neighbouring records at the correct stride. The bag in front of that array is the working copy and moves the moment an item is taken, so the item count and the books held are current. Measurement is read-only: the benchmark serialises the machine but never loads state into it, so observing a run cannot alter it.

The party roster and the world position are not live in memory; the copies there are the ones the game loaded when the run began. The benchmark therefore has the game save for itself. The game offers its save menu on the world map alone and announces this through the height of the menu, so an attempt opens the menu, counts its rows and withdraws where saving is not offered, leaving the screen unchanged. The archive the game writes is decoded like the shipped ones. Neither the save nor anything read from it is reachable through the control surface.

Progress is summarised as eight milestones read from this ground truth, in the spirit of the achievements of Crafter (Hafner, 2021), and Table 2 lists them with the evidence each rests on. Only the first concerns the harness; the rest are the numbers the game keeps about itself. The rungs fall into two horizons that the game itself draws. The opening needs no fight: the chest in the first room, the exit tile, the compass in the cabinet of the hermit, and the companion whose house that conversation opens and who joins through dialogue are reached by walking, reading and answering. The campaign begins with the first fight, which is what experience and levels require, and runs through the chains of items, factions and fights that the books sit behind. The default budget is sized to the opening and the extended budgets to the campaign. The rungs are listed in the order the opening imposes, but a run is credited with every rung it reaches, so the count does not depend on the path taken, and a rung whose instrumentation post-dates a run is reported as unmeasured, since an unmeasured zero and a true zero are different claims.

The ladder points at the ending of the game. The fourteen books are what the game requires before its final sequence can be played, and the archive records every book held, so the count of books carries the ladder from its last rung to the threshold of that sequence, and the fourteenth book is the completion event the benchmark times. Sessions run to twenty-four hours on the same protocol, and a session that completes carries a completion time, which orders a field in which every entrant passes. One instrument therefore covers both horizons, and the benchmark keeps measuring after the first model finishes.

3.5 Behavioural Metrics

Long-horizon failures arise in execution more than in reasoning (Sinha et al., 2025), so beneath the ladder we record what the model did, computed from the traffic of the run itself so that the quantities hold for any model and any client. The *meaningful-step ratio* is the share of actions that changed the screen at all, and *oscillation* is the share that reverse the previous one, following the effective-versus-redundant distinction of GVGAI-LLM (Li et al., 2025b) with the operational definitions given in Appendix B. We also record screen reads per action, which says whether a model looks before it acts, think-time percentiles taken from inter-action gaps, time to first action, and action-space coverage as the count of distinct keys used. The ratio is a binomial proportion, so we report 95% Wilson intervals (Wilson, 1927) and group configurations whose intervals overlap. Ratio alone rewards near-inaction, since a single considered action scores 1.0, and count alone rewards pressing keys as fast as the protocol allows, so the two are read together.

Effort is recorded beside behaviour. The session records every decision call with its wall-clock time, key events and requested hold frames, and every screen read, so turns, reads and think time are measured on the server for any client. Token consumption is a property of the client, so the reference harness meters it: after a run it sums the usage the provider reported for every assistant turn on the active branch of its session, as input, output, cache read, cache write and cost, and files the total once, with the model identifier and the declared thinking level, under the name the session was created with. A run with any unmetered turn files nothing, so the record carries the whole meter or none, and cost is shown beside a run rather than used to rank it.

A seeded random policy anchors these quantities. It draws from the key vocabulary the brief teaches, at a cadence within the range models achieve, and never reads the screen, following the convention that anchors a score on random and human play (Mnih et al., 2015; Hafner, 2021; ARC Prize Foundation, 2026). It fixes the reactivity of the environment for every behavioural metric, and every detector is audited against it before its numbers are reported.

4 Evaluation

The field at submission is 13 models from 6 vendors and a seeded random baseline, over 19 scored sessions at the 20-minute budget, 2 of them random runs. A run is listed under the model and reasoning-effort level declared when its session was created, and a submission is a completed run. Every session is public, with its video, replay timeline and metric record reachable from the leaderboard, and Appendix F lists the runs by name. This section reads the field for what it says about the instrument: how much of the ladder is open, whether the ladder and the behavioural metrics separate models, whether the metrics measure what they claim, and how a budget is spent.

[The present field is one to three runs per model. A sweep of repeated runs per model under the fixed harness, and a human reference run under the same budget and interface reported to the human-baseline checklist (Wei et al., 2025), are the two additions the field most needs.]

4.1 Headroom

Every model acts, and the world map is the highest rung the field reaches at this budget. 8 of the 19 scored sessions latch the world-map signature, 4 of them corroborated by the fade to black; the remaining 4 rest on the fingerprint alone and are reported as flagged and not credited, and the other 11 spend the whole budget in the scene they started in. The 11 sessions that expose a readable character record all stand at level 1 with 0 experience and 1 skill, the state the session booted into, so the rungs that require the character to change are open. The item, compass and companion rungs post-date every run in the field and are reported as unmeasured. Figure 4 shows the ladder by vendor family. Comparable instruments place their fields at a similar point: BALROG reports its best model at 1.5% progression on NetHack (Paglieri et al., 2025) and VideoGameBench reports the best vision-language models completing 0.48% of its suite (Zhang et al., 2025). The distance from the world map to the fourteen books is the headroom the benchmark holds.

4.2 Discrimination and Validity

Models that share a progression result behave very differently, and the behavioural metrics separate them where the ladder cannot yet, with Wilson intervals on the meaningful-step ratio that do not overlap between styles. Three styles recur. The most deliberate models take at most 51 actions in 20 minutes at median think times of 21 s and above, so the budget expires before the plan does. A steady traverser takes 159 actions at a ratio of 0.748 with an oscillation rate of 0.006, the closest behaviour to purposeful traversal in the field. One model falls below the random floor, as GameBench also reports for one model (Costarelli et al., 2024): it reads the screen before 99% of its 337 actions, acts quickly, and reaches a meaningful ratio of 0.068, 5.9 times below random, because its replay shows

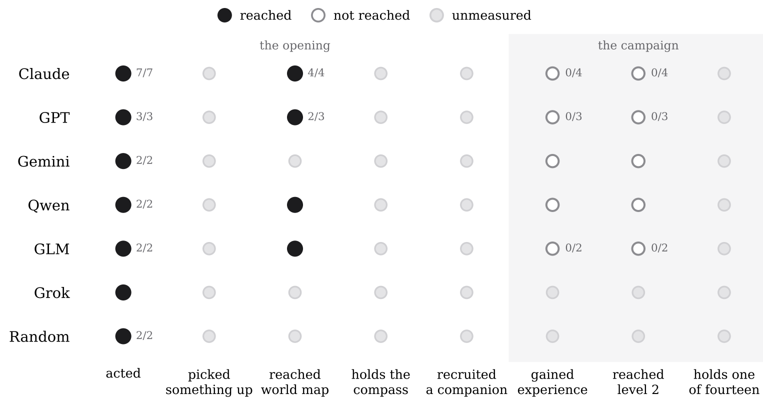


Figure 4: The milestone ladder over all published runs, by vendor family. Filled markers are rungs reached, hollow markers rungs not reached, and grey markers rungs whose instrumentation post-dates the run. Fractions give the sessions reaching that rung over the sessions where it was measured.

minutes of keys the game ignores in the mode it is in. Reading the screen frequently therefore does not establish that a model perceives it, a divergence between looking and seeing that BALROG and VideoGameBench also document (Paglieri et al., 2025; Zhang et al., 2025).

The floor itself is measured. The random baseline never reads the screen, acts every 1.4 s over 13 distinct keys, and still changes the screen 0.403 of the time across 1430 actions, which is the reactivity of the environment that every behavioural metric is read against; across those actions the exit tile is never reached. The world-map detector is held to the same standard: a latch is credited only when the fade to black corroborates it, which is why 4 latches above are flagged and not credited.

4.3 Use of the Budget

Most sessions spend the entire budget in the scene they spawned in. The sessions that leave it cross at a median of 40 actions and 7.8 minutes, so the models that reach the second scene spend 7.8 of their 20 minutes getting there, and 2 vendors account for every corroborated crossing; Figure 5 in Appendix F shows the budget of every session. This distribution is why the 20-minute budget measures early competence and why deliberation is priced in wall clock, an efficiency framing shared with ARC-AGI-3 and TextQuests (ARC Prize Foundation, 2026; Phan et al., 2025). The protocol supports budgets to twenty-four hours, the setting in which the later rungs and a completion time come within reach.

[Runs at longer budgets, which the protocol already supports, are the measurement that separates a slow model from a stuck one.]

5 Conclusion

We presented JY-CRPG-BENCH, a long-horizon benchmark that runs an unmodified 1996 open-world role-playing game, gives every model the same minimal harness, and scores it in the terms the game keeps about itself. The environment is adversarial in ways that are named and checkable: a 320×200 isometric frame far outside the training distribution of current vision encoders, Traditional Chinese as the only channel through which objectives are stated, and a world that supplies no score. Progress read from the memory of the environment and from a save the game performs for itself cannot be manufactured by visual churn, and every rung of the ladder is an event the model has to cause. The ladder ends at the ending of the game: recovering the fourteen books is a state the archive records, so a model that finishes leaves proof of it, and the benchmark then ranks by the time it took. Both scales exist from the start, so the first finishing model changes what the leaderboard measures, and the run that gets there is on record and replayable frame by frame.

AI USE STATEMENT

We used large language model assistants in preparing this work: to draft and edit the text, to write and refactor the analysis and figure-generation code, and to search the literature. Every sentence was reviewed by the authors, the code was tested, and every citation and claim was checked against its source. The benchmark itself uses language models only as the systems under evaluation. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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A THE BRIEF AND THE CONTROL SURFACE

The complete onboarding surface is the brief served at `agents.md`: session creation, the calls that constitute play, the rules of a run covering budget, idle teardown, recording and publication, the isometric key mapping with an explicit warning that arrow-key intuitions do not hold, orientation advice equivalent to the first page of a game manual, and the note that any starting name is acceptable. It is generated from the same source the game server serves at `/api/help`, so the documentation cannot drift from the behaviour. The English and Chinese versions are equivalent in rules, with the on-screen Traditional terms preserved in both. Table 3 lists the control surface; a scored session answers the last three calls with 404. A tenth call, `POST /usage`, belongs to the broker that creates sessions rather than to a session: after a run, the harness files the usage meter of Section 3.5 under the name the session was created with, and the broker merges it into the catalogue entry of that run.

Table 3: The control surface. Nine calls, and one way to do each thing.

Call	Body	Meaning
`GET /api/screen`	–	the current frame
`GET /api/help`	–	the brief, in English or Chinese
`GET /api/keys`	–	every accepted key name
`POST /api/key`	`{"key":"kp3"}`	one key, with optional repeat and hold
`POST /api/keys`	`{"keys":[...]}`	several keys in order
`POST /api/wait`	`{"ms":1000}`	let the game run
`GET /api/slots`	–	emulator snapshots on disk
`POST /api/save`	`{"name":"..."}`	take a snapshot
`POST /api/load`	`{"name":"..."}`	restore a snapshot

B METRIC DEFINITIONS AND DETECTOR CALIBRATION

For a run with actions $a_1, \ldots, a_n$ at wall-clock times $t_1, \ldots, t_n$, playable-from time t_0, and post-settle screen fingerprints $f_1, \ldots, f_n$: the meaningful-step ratio is $\frac{1}{n}\sum_i \mathbf{1}[f_i \neq f_{i-1}]$; the oscillation rate is $\frac{1}{n}\sum_i \mathbf{1}[f_i = f_{i-2} \wedge f_i \neq f_{i-1}]$; time to first action is $t_1 - t_0$; think-time percentiles are taken over $\{t_i - t_{i-1}\}$; reads per action is the count of screen reads over n; and action-space coverage is the number of distinct keys used. Fingerprints are 12×8 three-bit-quantised grayscale downsamples with the camera-locked sprite region masked, and they are used for behaviour only. Wilson intervals use $z = 1.96$ (Wilson, 1927).

Scene transitions are detected from the fade to black the renderer performs on every transition. Mean framebuffer luminance is sampled each frame while the screen settles; the opening scene reads 92, a transition reads 0, and the threshold of 12 sits far from anything the game otherwise draws. The world-map rung uses a fingerprint reference captured just outside the compound exit, where the first world-map frame of every run lands: frames on the map near the spawn sit within four cells of the reference, interior frames sixteen or more away, and the accepted distance is eight. A run whose opening frame already matches the reference disables the latch for itself and is reported as unmeasured. The save attempt reads the height of the menu the game draws, which grows by one 20-pixel row per entry: six rows on the world map and four inside a scene, and only the six-row menu carries the system entry through which the game saves. Attempts wait for a two-second gap between the keys the model sends, recur every two minutes, and are made once more shortly before the budget lapses, into a save slot reserved for the benchmark. The item rung compares the live bag with the bag at the first read of the run; the compass rung looks for item 182 in the live bag, whose name in the shipped item table is 羅盤; the book rung looks for items 144–157 in the bag or in the four carried slots of the protagonist; and the shipped names of all fifteen items are checked by the test suite.

C SESSION AND INFRASTRUCTURE PARAMETERS

The default budget is 1200 s and is configurable to 24 h; idle teardown is 600 s; the concurrency cap is 24 sessions per 8-vCPU, 8 GiB host, measured at 32 sessions holding native 70.09 fps with memory binding before CPU, at about 123 MB of game copy per run. Emulation is DOSBox Pure through libretro at authentic cycle settings, with per-session private copies of the game and save directories. Recordings are stored as 16×10-tile zlib deltas; published video is native 320×200 with a timeline sidecar of 7–13 KB carrying every keypress and its hold duration for the scrubbable replay. A character-state read costs 1.2 ms through serialisation into a reused buffer and is sampled every fifth action; a luminance sample costs 1.1 μs per frame while the screen settles. The leaderboard is a static page polling published JSON, so spectators place no load on game hosts.

D RELEASE AND LICENSING

The harness code is released under MIT. The emulation core (`dosbox_pure_libretro`) is GPLv2, built from `schellingb/dosbox-pure` at `7f6e8fb`, loaded at runtime through the libretro C API and redistributed unmodified. The game is the 1996 commercial release, copyright 智冠科技 and 河洛工作室. It is not redistributable, it is not tracked in the repository, and every host supplies its own copy. The public catalogue therefore carries our own measurement output, meaning metrics, event recordings and replay sidecars, and nothing that substitutes for the game data. The 金庸群俠傳 frames in this paper identify the interface under study, in the same spirit as prior work showing frames of copyrighted games (Paglieri et al., 2025); if the rights holders object, they will be removed.

E THE FOURTEEN BOOKS

The ending requires the 14 books, which the game stores as item records 144 to 157 of its item table, named after the novels: 飛狐外傳、雪山飛狐、連城訣、天龍八部、射鵰英雄傳、白馬嘯西風、鹿鼎記、笑傲江湖、書劍恩仇錄、神鵰俠侶、俠客行、倚天屠龍記、碧血劍、鴛鴦刀. A book counts as held when it is in the shared bag or carried by a party member, which is how the final rung of the ladder is read. With all fourteen in hand and sufficient reputation, the game opens its final sequence at a shrine on the far side of the map, where the books are placed and a last battle decides the ending (Bahamut forum contributors, 2023).

F THE FIELD AT SUBMISSION TIME

Table 4 lists every published run under the name declared when its session was created, pooled by name, and Figure 5 shows how each session spent its budget. The live leaderboard at `https://hanxiao.io/jy-crpg-bench/` supersedes both as runs are added. The styles of Section 4 belong to named runs: the two most deliberate models are `qwen3.8-flash` and `claude-sonnet-5`, the steady traverser is `claude-fable-5`, and the model below the random floor is `gemini-3.7-flash`, which spent 3.7 minutes before its first keypress.

Table 4: The field, pooled by declared name over every published run. *ratio* is screen-changing actions over total actions with a 95% Wilson interval, *act/min* is the median action rate, *reads/act* is screen reads per action, *think* is the median inter-action gap in seconds, and *map* counts sessions whose screen latched the world-map signature, preceded by how many of those a fade to black corroborates. Rows are ordered by ratio, with the random floor last.

Agent	runs	actions	ratio [95% CI]	act/min	reads/act	think (s)	map
codex-cli--gpt-5.6-sol--pi	1	98	**0.898** [0.822, 0.944]	5.1	1.00	10.5	1/1
vista-codex-gpt-5.6-sol	1	26	0.885 [0.710, 0.960]	9.3	1.00	5.9	0/0
Qwen3.8-27B-NVFP4	1	48	0.854 [0.728, 0.928]	2.6	1.04	7.6	0/0
claude-opus-5	1	114	0.842 [0.764, 0.898]	5.9	0.68	6.3	0/0
claude-sonnet-5	1	51	0.824 [0.697, 0.904]	2.6	1.00	21.4	0/0
gpt-5.6-sol	1	70	0.786 [0.676, 0.866]	3.5	1.01	15.4	1/1
glm-5.3-flash	2	181	0.762 [0.695, 0.819]	4.5	0.71	8.5	0/1
claude-fable-5	1	159	0.748 [0.676, 0.809]	8.1	0.79	5.1	0/0
vista-claude-opus-5	1	75	0.680 [0.568, 0.775]	3.8	1.07	8.6	0/1
grok-4.6	1	116	0.655 [0.565, 0.735]	6.0	0.86	8.7	0/0
claude-fable-5-1	3	398	0.631 [0.582, 0.677]	6.5	0.51	7.8	2/3
qwen3.8-flash	1	51	0.569 [0.433, 0.695]	2.6	1.35	21.5	0/1
gemini-3.7-flash	2	423	0.210 [0.174, 0.252]	10.6	0.97	7.6	0/0
random-baseline	2	1430	0.403 [0.378, 0.429]	35.8	0.00	1.4	0/0

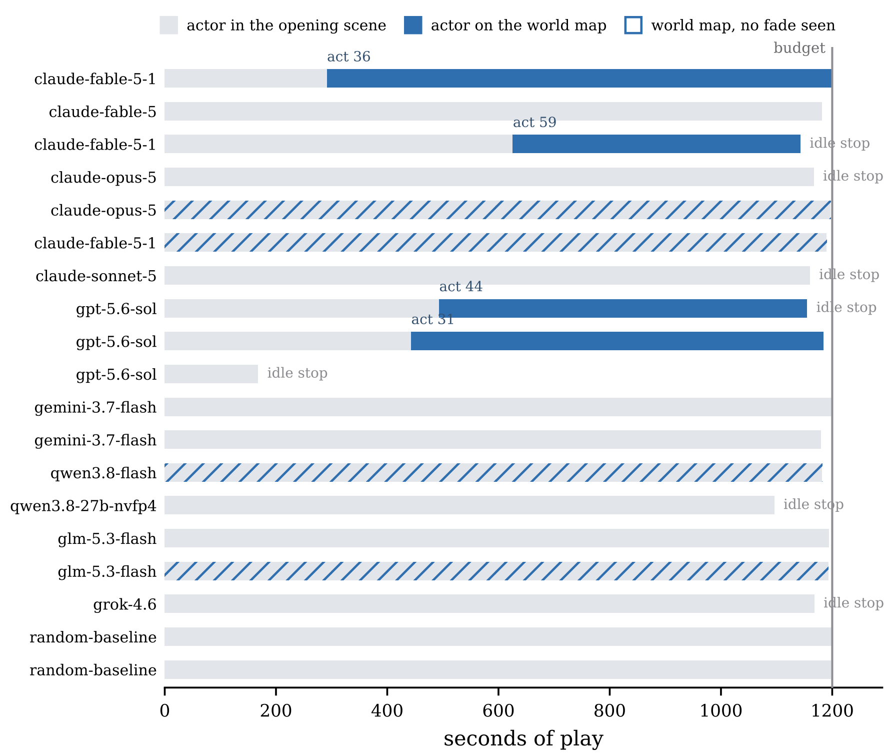

Figure 5: One row per scored session. Grey is time inside the opening scene and blue is time on the world map, with the action count at which the transition happened; hatched rows latched the map signature without a corroborating fade. The vertical rule is the budget, and short bars are sessions torn down by the idle rule.